

Dibakar Roy Chowdhury

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Professor of Physics,

Nationality: Indian

Ecole Centrale School of Engineering,

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EDUCATION

- **PhD (2003 - 2008):** Technical University of Darmstadt, Darmstadt, Germany
Thesis title: Experimental study and modeling of AC characteristics of Resonant Tunneling Diodes (RTD)
Advisors: Dr. Michael Feiginov and Prof. Peter Meissner
- **M. Tech (Solid state technology), 2001-2003:** Indian Institute of Technology, Madras, Class rank: 1 (cgpa: 8.74/10)
M. Tech Project Title: Fabrication and characterization of Heterostructure Barrier Varactors and Schottky diodes for THz applications.
Advisors: Prof. Hans. L. Hartnagel and Prof. Amitava Dasgupta
- **MSc (Physics, specialization in solid state physics), 1998-2000:** Rani Durgavati Vishwavidyalaya, Jabalpur, India, University rank: 5 (75.75%)

RESEARCH INTERESTS

- Metamaterials, Plasmonics
- Ultrafast Spectroscopy and Terahertz Spectroscopy
- Ultrafast and non-linear electronic devices
- Micro- and nano- structure fabrication

RESEARCH EXPERIENCE

- **Professor,** Mahindra University, India
October 2018 –
 - Metamaterials, Plasmonics
 - Terahertz Spectroscopy
 - Micro- Nano- Fabrication
- **Associate Professor,** Mahindra Ecole Centrale, India
January 2015 – October 2018
 - Terahertz
 - Metamaterials
 - Plasmonics,

- **Scientist - level B (equivalent to Senior Lecturer)**, Australian National University, Canberra, Australia August 2013 – January 2015
 - Plasmonics
 - Metamaterials
- **Research associate**, National University of Singapore, Singapore March 2013 – July, 2013
 - Metamaterials
- **Postdoctoral research associate**, Center of Integrated Nanotechnologies (CINT), Los Alamos National Laboratory, NM, USA June 2009 – March 2013
 - Active and non-linear metamaterials
 - Terahertz Science & Technology
 - Absorbers and Ultrafast phenomena
- **Postdoctoral research associate**, Center for Nanointegration Duisburg-Essen (CeNIDE), University of Duisburg-Essen, Duisburg, Germany Apr 2008 - Apr 2009
 - Synthesis of nanostructures
 - Electrical and structural characterization
- **Graduate Research Assistant**, Faculty of Electrical Engineering, Technical University of Darmstadt, Darmstadt, Germany Aug, 2003 – Mar, 2008
 - Ultrafast electronic devices RTD, Schottky diode etc. for THz frequencies
 - Fabrication and electrical characterizations of micro- nano- structures

RESEARCH SKILLS

Summary of skills:

- Electromagnetics and semiconductor device physics
- **Simulation techniques:** Static and dynamic simulations of novel semiconductor devices (RTD, HBV etc.), CST microwave studio for electromagnetic simulations
- **Software skills:** Matlab, mathematica, origin, Ledit
- **Experimental skills:**
 - **Micro- and Nano- fabrication:** Photolithography, Electron Beam Lithography (EBL), wet and dry chemical etching, electron beam deposition of metals, thermal deposition of metals and organics, oxygen plasma etching and cleaning, rapid thermal processing, ellipsometry and profilometric techniques, Scanning Electron Microscope (SEM)
 - **Electrical Measurements:** Parameter analyzer, Vector network analyzer (VNA), Spectrum Analyzer (SA), etc.

- **Ultrafast measurements:** Terahertz time domain spectroscopy (THz-TDS), Optical Pump Terahertz Probe (OPTP) measurements

TEACHING EXPERIENCE

Courses Taught:

- Modern Physics
- Solid State Devices
- Quantum Mechanics
- Electromagnetism and Optics
- Electronics
- Semiconductor Devices

Laboratories conducted:

- Physics-II (Electromagnetism and Optics) Laboratory for B Tech (2015, 2016)
- Electronics Laboratory (2015)
- Graduate Laboratory course in basic Electronics in University of Duisburg-Essen, Germany (2008-2009)
- Tutor for high frequency electronics, Technical University of Darmstadt, Germany (2004-2005)

LABORATORY DEVELOPMENT

- Developed Micro fabrication facility comprising of optical lithography, evaporation set up etc in Mahindra Ecole Centrale, India (INR 1 Crore +)
- Developed High frequency electronics laboratory in collaboration with Tektronix in Mahindra Ecole Centrale, India (INR 1.2 Crore +)
- Developed THz laboratory (INR 1 Crore)
- Developed Physics-II Teaching Laboratory in Mahindra Ecole Centrale for B. Tech 2nd year students: Experiments in Electricity, Magnetism and Optics (INR 35 Lakh +)

STUDENTS MENTORED

PhD

- S Jagan Mohan, PhD, Thesis Title “Exploring the Near Field Coupling and Ultrafast Switching in Terahertz Metamaterials” (2020)
- Sabyasachi Banerjee, PhD, Thesis Title “Development of Metamaterial based Terahertz Sensing Devices using the approach of surface waves and resonators spanning different design styles” (2022)
- Subhajit Karmakar, PhD, Thesis Title “Studies on non-radiating metamaterials for Terahertz photonics”, (2022)

- Deepak Kumar, PhD, Thesis Title “Metamaterial inspired multi-functional structures for terahertz applications” (2022)
- Soumya Jyothi Mallick, PhD, Thesis Title “Dynamically Reconfigurable Near-field Coupled Terahertz Metasurfaces” (2024)

B.Tech

- Surya Pranav, B Tech Final year Thesis,” Graphene based terahertz metamaterials”, Mahindra Ecole Centrale, Hyderabad (2020)
- Ch. Amith, B Tech Final Year Thesis, “Coupling Slot Antenna with Metamaterials”, Mahindra Ecole Centrale, Hyderabad (2019)
- Nakka Lok Abhishikht, B Tech Final year Thesis, Title “Controlling extraordinary transmission through hole arrays using sub wavelength arrays”, Mahindra Ecole Centrale, Hyderabad (2018)
- Eppa Akhilesh Reddy, B Tech Final year Thesis, Title “Terahertz Metamaterials based Thin Film sensors with ground plane” Mahindra Ecole Centrale, Hyderabad (2018)
- Shravan Kumar, B. Tech final year Thesis Title “Terahertz propagation through conducting and non-conducting medium” in collaboration with IIT Guwahati, Guwahati, India (2016)
- Levon Altaveyon, Thesis Title “Evaluation of PMMA as gate dielectric for field effect transistors” University of Duisburg-Essen, Germany (2009)

M.S.

- Shreeya Rane, MS Thesis, Title “Terahertz metasurfaces for anomalous refraction”
- Sowmya Purankar, Thesis Title “Metamaterials based Microwave electromagnetic absorbers” Master’s project (2016)
- Vyshnavi Reddy, Thesis Title “Thin film sensing with Metamaterials” Master’s project (2016)

PhD STUDENTS UNDER SUPERVISION

- Sukhvinder Kaur, PhD, Project “Terahertz Metamaterials” (started in 2019)
- Shreeya Rane, PhD, Project Title “Sub wavelength Terahertz grating” (started in 2020)
- Soumya Jyothi Mallick, PhD, Project Title “Nonlinear Terahertz metasurfaces” (started in 2020)
- Nityananda Acharyya, PhD, project “Terahertz spintronics” (started in 2021)
- Indu Krishna KN, PhD, Project Title “Terahertz metamaterials” (started in 2020)
- Vaishnavi Sajeev, PhD, Project Title “Terahertz plasmonics” (started in 2020)

POST DOCTORALS SUPERVISIONS

- Dr. Mangababu Akkibonoina (PhD Hyderabad University, Hyderabad, India), Project Title “Materials for Terahertz domain” (2023 - 2024),
- Dr. Gopal Kulkarni (PhD Shivaji University, Kolhapur, India), Project Title “Topological photonic crystals” (2023 - 2024),
- Dr. Koijam Monika Devi (PhD IIT Guwahati, Guwahati, India), Project Title “Terahertz Topological photonics” (2020 - 2022), now with Lille University, France
- Dr. Shahjahan S (PhD Jamia Milia, New Delhi, India), Project Title “Fabrication of Terahertz Metamaterials” (2016-2018)

AWARDS AND DISTINCTIONS

- Cleared **CSIR** (Council for Scientific and Industrial Research) - **NET** (National Eligibility Test) examination for junior research fellowship and for eligibility of lecturership in India (2000)
- **Graduate Aptitude Test in Engineering (GATE)** [jointly organized by “Indian Institute of Technology” and “Indian Institute of Science”]: All India rank – 19 (2001)
- Cleared **GRE** and **TOEFL** examinations (2002)
- **Institute Gold medal (2003)**: For securing top position in the order of merit in the branch of solid state technology at the Master of Technology examination from Indian Institute of Technology, Madras, India
- **Institute fellowship (2001 - 2002)**: In order to pursue Master of Technology from Indian Institute of Technology, Madras, India
- **DAAD Fellowship (2002 - 2003)**: For standing first in the class of Solid state technology to pursue the Master’s thesis from the selective technical universities of Germany
- **Early Career Research Award (2016)**: Department of Science and Technology (DST), Government of India
- **Received Excellent grade (2020)**: For SERB project EMR/2015/001339
- **Fellow of IETE (2016)**
- **Fellow of OSI (2017)**
- **OSA Senior member (2021)**
- **Fellow of Institute of Physics, FInstP (UK), (2022)**
- **Fellow of Telangana Academy of Sciences, (FTAS) (2023)**
- **Recipient of SERB-STAR (Science and Technology Award for Research) award (2023)**

REVIEWER

- Optics Letters (OL)
- Applied Optics (AO)
- RSC Advances (RSC-A)
- Journal of Optics (JOPT)

- Advanced Materials (AM)
- Applied Physics Letters (APL)
- Current Applied Physics (CAP)
- Journal of NanoPhotonics (JNP)
- Journal of Applied Physics (JAP)
- Smart Materials Structures (SMS)
- Progress In Electromagnetic Research (PIER)
- Scientific Report: Nature group of Journals (SR)
- IEEE Photonics Technology Letters (IEEE-PTL)
- Journal of Physics D: Applied Physics (JPHYSD)
- Journal of Optical Society of America B (JOSA-B)
- IEEE Transactions on NanoTechnology (TNANO)
- IEEE Journal of Lightwave Technology (IEEE-JLT)
- IEEE Journal of the Electron Devices Society (J-EDS)
- American Chemical Society Photonics (ACS-Photonics)
- IEEE Antennas and Propagation Magazine (IEEE-APM)
- Journal of Infrared Millimeter and Terahertz Waves (JIMT)
- Journal of ElectroMagnetic Waves and Applications (JEMWA)
- International Journal of Electronics and Communications (IJEC)
- IEEE Journal of Selected Topics in Quantum Electronics (JSTQE)
- Photonics and Nanostructures-Fundamentals and Applications (PNFA)
- More

MEMBERSHIP TO PROFESSIONAL BODIES

- Fellow Institute of Physics, (FInstP, UK)
- Fellow Telangana Academy of Sciences (FTAS)
- Fellow Institute of Electronics and Telecommunication Engineers (IETE)
- Senior Member Optical Society of America (OSA)
- Fellow Optical Society of India (OSI)

PATENTS

1. *Multilayered (Super-lattice) Metamaterials based Terahertz Spintronic Devices and its method of fabrication*, **Dibakar Roy Chowdhury**, Subhajit Karmakar, Deepak Kumar, and Ravi Varshney, Indian Patent, **Patent number: 453039** (2021)
2. *A Topological Waveguide Device and its method for Dispersion less Slow Light Transmission at Terahertz (THz) Frequencies*, **Dibakar Roy Chowdhury**, and Koijam Monika Devi, Indian Patent, **Patent number: 483683** (2021)
3. *Stacked Fano Metamaterials based thin film sensors*, **Dibakar Roy Chowdhury**, Subhajit Karmakar, Ravi Varshney, Sabyasachi Banerjee, Deepak Kumar, **Patent number: 491812**, Application Number: 202041003489 (2020)

4. A Topological Waveguide Device for sustaining dynamically tunable propagation of electromagnetic waves at microwave frequencies, **Dibakar Roy Chowdhury**, and Koijam Monika Devi and Sambhu Jana, Application number: 202241068284 (2022)
5. *Evanescent Order driven deep sub-wavelength particle detector employing all-dielectric terahertz metagrating (sub-wavelength grating)*, **Dibakar Roy Chowdhury**, Shreeya Rane, and Shriganesh Prabhu, Application number: 202141029457 (2021)
6. *Hand Held Metasurface based Microbe Detection Device (H2M2D2)*, **Dibakar Roy Chowdhury** and Sabyasachi Banerjee, Indian Patent Applied, Application Number: 202041026401 (2020)

PUBLICATIONS

1. *Multi-stacked polarization insensitive broadband terahertz metamaterial*, Bhagwat Singh Chouhan, Nityananda Acharyya, Anuraj Panwar, **Dibakar Roy Chowdhury**, and Gagan Kumar, Journal of Applied Physics, **135**, 173103(2024)
2. *Tailoring terahertz surface plasmon resonances through near-field coupling in multiple holes-based arrays*, Vaishnavi Sajeev, Shreeya Rane, Nityananda Acharyya, and **Dibakar Roy Chowdhury**, Results in Optics, **16**, 100678 (2024)
3. *Resonant modes in Topological Ring Resonator (TRR)*, Geetanjali Jena, Gopal Kulkarni, Ravi Varshney, and **Dibakar Roy Chowdhury**, Accepted, Journal of Physics D: Applied Physics **57**, 305101 (2024)
4. *Exceptional point THz sensing with metasurfaces*, Indu Krishna K N, and **Dibakar Roy Chowdhury**, New Journal of Physics, **26**, 053033 (2024)
5. *Electrical and Electrochemical Investigations on Copper Oxide-based Monolayer and Bilayer Memristive Devices*, Somnath S. Kundale; Akash N. Kurade; Pradnya P. Patil; Deepali S. Shivade; Swapnil R. Patil; Kasturi A. Rokade; Krantiveer V. More; **Dibakar Roy Chowdhury**; Rajanish K. Kamat; Tukaram D. Dongale, ACS applied electronic materials, (2024)
6. *Physics and applications of terahertz metagrating*, Shreeya Rane, Shriganesh S Prabhu, and **Dibakar Roy Chowdhury**, Opto Electronic Sciences, accepted, review article (2024)
7. *Magnetically controlled Toroidal metasurfaces*, Nityananda Acharyya, Soumyajyoti Mallick, Shreeya Rane, Mousumi Upadhyay, and **Dibakar Roy Chowdhury**, Advanced Optical Materials, 2303045 (2024)
8. *Deep sub-wavelength dipole cavities for terahertz thin film sensing*, Vaishnavi Sajeev, Mangababu A, Nityananda Acharyya, Indu K N, and **Dibakar Roy Chowdhury**, Optics Communications, accepted (2024)
9. *Fano resonance-based CNT detection using THz MM*, Shreeya Rane, Soumya Jyoti Mallick, Saloni Sharma, Nityananda Acharyya, Bipin Kumar Gupta, and **Dibakar Roy Chowdhury**, ACS Applied Optical Materials, **1 (12)**, 2004 (2023)
10. *Electronically triggered tunable terahertz signal observed in LPCVD Grown single layer Graphene*, Saloni Sharma, Shreeya Rane, Z A Ansari, **Dibakar Roy Chowdhury**, and Bipin Kumar Gupta, Journal of Materials Chemistry C, **11**, 14805 (2023)

11. *Role of Losses and Coupling in Bright-Dark Metasurfaces mimicking non-Hermitian parity-time symmetric systems*, Indu K N, Ajinkya Punjal, Naga Deepthi K N, Shriganesh S Prabhu and **Dibakar Roy Chowdhury**, Journal of Applied Physics, **134**, 163103 (2023)
12. *Tunable Broadband Terahertz Absorption and Shielding of Bioderived Graphitic Carbon Microspheres*, Gopal Kulkarni, Shreeya Rane, Nityananda Acharyya, **Dibakar Roy Chowdhury**, and Vijaya Rao, Ceramics International, **49**, 39441 (2023)
13. *High Performing TiO₂ Flower-like Nanostructure Based on Flexible MWCNTs for Dual-band Terahertz Absorption*, Guruvendra Singh, Subhash Nimanpore, Nityananda Acharyya, Shreeya Rane, **Dibakar Roy Chowdhury**, Bhanu Singh, Jai s Tawale, Rina Sharma, and Mukesh Jewariya, Journal of Materials Chemistry C, **11**, 14199 (2023)
14. *Realizing dynamically tunable topologically protected electromagnetic wave propagations*, Sambhu Jana, Koijam Monika Devi, Gopal Kulkarni, Soumyajyothi Mallick, and **Dibakar Roy Chowdhury**, New Journal of Physics, **25**, 093023 (2023)
15. *Room temperature-grown tungsten oxide hybridized dipole cavities to realize thermally tunable terahertz surface plasmons*, Arun Jana, Atul Khot, Shreeya Rane, Vaishnavi Sajeev, T K Dongale, T G Kim, and **Dibakar Roy Chowdhury**, Optical Materials, **143**, 114274 (2023)
16. *Design, optimization and characterization of Evanescent orders in Terahertz meta gratings*, Shreeya Rane, Ajinkya Punjal, Shriganesh Prabhu and **Dibakar Roy Chowdhury**, Optics Continuum, **2**, 1996 (2023)
17. *Reduced Graphene Oxide detection with hole arrays*, Vaishnavi Sajeev, Shreeya Rane, Debal Ghosh, Nityananda Acharyya, Palash Roy Chowdhury, Arnab Mukherjee, and **Dibakar Roy Chowdhury**, Scientific Reports, **13**, 12374 (2023)
18. *Dynamic Terahertz wave propagation through Al/Ni based spin valve structures*, Nityananda Acharyya, Shreeya Rane, Vaishnavi Sajeev, Subhajit Karmakar, and **Dibakar Roy Chowdhury**, Journal of Applied Physics, **134**, 033102 (2023)
19. *Accessing dual toroidal modes through polarization sensitive resonance hybridizations in terahertz metasurfaces*, Soumya Jyothi Mallick, Shreeya Rane, Nityananda Acharyya, and **Dibakar Roy Chowdhury**, New Journal of Physics, **25**, 053016 (2023)
20. *Fourier Transformed Terahertz Spectroscopy inspired detection of evanescent orders in all dielectric sub-wavelength grating*, Shreeya Rane, Ajinkya Punjal, Shriganesh Prabhu, and **Dibakar Roy Chowdhury**, IEEE Journal of Selected Topics in Quantum Electronics, **29**, 8500406 (2023)
21. *Broadband THz absorption using nanosheets of Bi₂Te₃ grown on a transparent conductor*, Animesh Pandey, Reena Yadav, Sabyasachi Banerjee, Sumeet Kumar, Subhash Nimanpore, Mukesh Jewariya, **Dibakar Roy Chowdhury**, N P Lalla, and Sudhir Husale, Journal of Physical Chemistry C, **11**, 1448 - 1456 (2023)
22. *Multi-mode hybridizations in surface plasmon resonances excited in terahertz dipole cavities*, Sukhvinder Kaur, Shreeya Rane, Ravendra K Varshney, and **Dibakar Roy Chowdhury**, JOSA B, **40**, 1251 (2023)
23. *Broadband terahertz linear cross polarization conversion in transmission mode using planar coupled metamaterials*, S Jagan Mohan, Rakesh Sarkar, Ajinkya Punjal,

- Deepa Ghindani, **Dibakar Roy Chowdhury**, Shriganesh Prabhu, and Gagan Kumar, *Journal of Applied Physics*, **133**, 123102 (2023)
24. *New Insights into APCVD Grown Monolayer MoS₂ using Time-Domain Terahertz Spectroscopy*, Saloni Sharma, Pooja Chauhan, Shreeya Rane, **Dibakar Roy Chowdhury**, and Bipin Kumar Gupta, *Scientific Reports*, **31**, 4146 (2023)
25. *Magnetotransport-induced non-contact terahertz detection of weak magnetic fields in optically-thin frequency-agile superlattice metasurfaces*, Subhajit Karmakar, Nityananda Achayya, Shreeya Rane, Ravendra Varshney, and **Dibakar Roy Chowdhury**, *Advanced Optical Materials*, **11**, 2202203 (2023); [Highlighted in Journal cover](#)
26. *Obstruction immune topological propagation*, Sambhu Jana, K Monika Devi, and **Dibakar Roy Chowdhury**, *Optics Communications*, **529**, 129111 (2023)
27. *Coulomb interaction mediated tuning of surface plasmon resonances in terahertz hole arrays*, Vaishnavi Sajeev, Arun Jana, Soumya Jyothi Mallick, Koijam Monika Devi, and **Dibakar Roy Chowdhury**, *Journal of Physics: Photonics*, **4**, 045001 (2022)
28. *Temperature tunable electromagnetically induced transparency in terahertz metasurface fabricated on ferroelectric platform*, Koijam Monika Devi, Arun Jana, Shreeya Rane, Palash Roy Choudhury, and **Dibakar Roy Chowdhury**, *Journal of Physics D: Applied Physics*, **55**, 495103 (2022)
29. *Terahertz characterizations of solution-processed Ni-doped, Cu-doped, and undoped ZnO nanoparticles*, Arun Jana, Tapas Das, Shreeya Rane, Koijam Monika Devi, Asim Guchhait, and **Dibakar Roy Chowdhury**, *Advances in Natural Sciences: Nanosciences and Nanotechnology*, **13**, 045009 (2022)
30. *Resonant Toroidal metasurface as a platform for thin film and biomaterial detection*, Sukhvinder Kaur, Arun Jana, Subhajit Karmakar, Ravendra Varshney, and **Dibakar Roy Chowdhury**, *Applied Optics*, **61**, 9020 (2022)
31. *Polarization independent dynamic modulation of slow light characteristics via electromagnetically induced transparency in terahertz metasurfaces*, Koijam Monika Devi, Arun Jana, Ajinkya Punjal, Nityananda Acharya, Shriganesh Prabhu, and **Dibakar Roy Chowdhury**, *New Journal of Physics*, **24**, 093004 (2022)
32. *Simultaneous Excitation of odd and even order resonances in plasmonic metasurfaces*, Soumya Jyothi Mallick, Arun Jana, Nityananda Acharyya, and **Dibakar Roy Chowdhury**, *Applied Optics*, **61**, 7435 (2022)
33. *Study of birefringence inside nanocrystalline Zinc Oxide thin films using terahertz spectroscopy*, Rahul Godiwal, Subhash Nimanpore, Guruvandra Singh, Amit Kumar Gangwar, Ajay Verma, **Dibakar Roy Chowdhury**, Pretam Singh, and Mukesh Jewariya, *Optical materials*, **133**, 112962 (2022)
34. *Magneto spectroscopy of terahertz surface plasmons in subwavelength perforated superlattice thin-films*, Subhajit Karmakar, Deepak Kumar, Ravendra Varshney, and **Dibakar Roy Chowdhury**, *Journal of Applied Physics*, **131**, 223102 (2022), [Cover page highlight/ feature article](#)
35. *Broadside coupling enabled insulator to metal transition in Vanadium-dioxide based multi-layer terahertz metasurfaces*, Soumyajyoti Mallick, and **Dibakar Roy Chowdhury**, *Euro Physics Letters*, **138**, 55001 (2022)
36. *Role of Unit cell defects in terahertz topological ring resonators*, Sambhu Jana, K Monika Devi, and **Dibakar Roy Chowdhury**, *JOSA B*, **39**, 1590 (2022)

37. *External bias dependent dynamic Terahertz propagation through BiFeO₃ film*, Arun Jana, Shreeya Rane, Palash Roy Choudhury, and **Dibakar Roy Chowdhury**, *Nanotechnology*, **33**, 325202 (2022)
38. *Graphene and metal hybridized terahertz metasurfaces toward tunable plasmon-induced transparency effects*, Indu Krishna K N, Monika Koijam Devi, and **Dibakar Roy Chowdhury**, *Current Applied Physics*, **39**, 158 (2022)
39. *Phase Engineered Hybrid Terahertz Metasurfaces*, Sukhvinder Kaur, Subhajit Karmakar, Arun Jana, Shreeya Rane, Ravendra Varshney, and **Dibakar Roy Chowdhury**, *iScience*, **25**, 104024 (2022)
40. *Broadband Terahertz characterization of Graphene Oxide films fabricated on flexible substrates*, Shreeya Rane, Avinash Kothuru, Arun Jana, Koijam Monika Devi, Sanket Goel, Shriganesh Prabhu, **Dibakar Roy Chowdhury**, *Optical Materials*, **125**, 112045 (2022)
41. *Photo induced dynamic tailoring of near-field coupled terahertz metasurfaces and its effect on Coulomb parameters*, Deepak Kumar, Manoj Gupta, Yogesh Shrivastava, Koijam Monika Devi, Ranjan Kumar, and **Dibakar Roy Chowdhury**, *Journal of Optics*, **24**, 045101 (2022)
42. *Effect of asymmetry on topological photonic crystals comprising of dielectric rod structures*, Sambhu Jana, K Monika Devi, and **Dibakar Roy Chowdhury**, *Optics Communications*, **505**, 127589 (2022)
43. *Magnetotransport modulation of Fano resonances in optically thin terahertz superlattice metasurfaces*, Subhajit Karmakar, Ravendra Varshney, and **Dibakar Roy Chowdhury**, *Journal of Physics D: Applied Physics*, **55**, 135109 (2021)
44. *Demonstration of toroidal metasurfaces through near-field coupling of bright mode resonators*, Soumyajyoti Mallick, Nitin Chourasia, Rakesh Singh, and **Dibakar Roy Chowdhury**, *Applied Physics Express*, **15**, 012005 (2021)
45. *Hybridization of Dark resonating states in terahertz metasurfaces*, Sukhvinder Kaur, Subhajit Karmakar, Arun Jana, Ravendra K Varshney, and **Dibakar Roy Chowdhury**, *Journal of Applied Physics*, **130**, 243101 (2021); [Cover page highlight/ feature article](#)
46. *Ultrasensitive Terahertz sensing with broadside coupled polarization insensitive graphene metamaterial cavities*, Sukhvinder Kaur, Subhajit Karmakar, K. Monika Devi, Ravendra K. Varshney, and **Dibakar Roy Chowdhury**, *Optik*, **248**, 168073 (2021)
47. *Investigation of dynamical optical study of Bi₂Te₃ topological insulators thin film based on MWCNT flexible paper using terahertz spectroscopy*. Subhash Nimanpore, Animesh Pandey, G Singh, B P Singh, **Dibakar Roy Chowdhury**, Y U Jeong, Rina Sharma, Sudhir Husale, and Mukesh Jewariya, *Optical materials*, **121**, 111490 (2021)
48. *Topological Edge states in an all dielectric terahertz photonic crystals*, Monika Koijam Devi, Sambhu Jana, and **Dibakar Roy Chowdhury**, *Optical Materials Express*, **11**, 2445 (2021)
49. *Elemental, Optical, and Time-Domain Terahertz Spectroscopy Studies on Methyl p-Hydroxybenzoate Single Crystal for THz Applications*, Debabrata Nayak, N Vijayan, Manju Kumari, Pargam Vashishtha, Sudha Yadav, Mukesh Jewariya, **Dibakar Roy Chowdhury**, Govind Gupta, and R. P. Pant, *Journal of Electronic Materials*, **50**, 6121 (2021)

50. *Dynamically tunable slow light characteristics in graphene-based Terahertz metasurfaces*, Deepak Kumar, Monika Koijjam Devi, Ranjan Kumar, and **Dibakar Roy Chowdhury**, Optics Communications, **491**, 126949 (2021)
51. *Magnetic wires: transverse magnetism in one-dimensional plasmonic system*, Subhajit Karmakar, Deepak Kumar, Bishnu P Pal, Ravendra Varshney, and **Dibakar Roy Chowdhury**, Optics Letters, **46**, 1365 (2021)
52. *Modulating Extraordinary Terahertz Transmissions in Multilayer Plasmonic Metasurfaces*, Sabyasachi Banerjee, Nakka Lok Abhishikth, Subhajit Karmakar, Deepak Kumar, Shreeya Rane, Sanket Goel, Abul K Azad, **Dibakar Roy Chowdhury**, Journal of Optics, **22**, 125101 (2020)
53. *Thin Film Sensing with Terahertz Metamaterials*, **Dibakar Roy Chowdhury**, Parama Pal, and Bishnu Pal, Asian Journal of Physics, **29**, 891-906 (2020)
54. *Growth and Characterization of Single Crystals of L-Histidine Hydrochloride Monohydrate for Nonlinear Optical Applications*, S Yadav, M Kumar, D Nayak, S Banerjee, N Khan, S Nimanpure, G Moona, R Sharma, B K. Sharma, **D. Roy Chowdhury**, N. Vijayan, and M Jewariya, Journal of Electronic Materials, **49**, 7502 (2020)
55. *Lattice Induced Plasmon Hybridization in Metamaterials*, Subhajit Karmakar, Deepak Kumar, Ravendra Varshney, and **Dibakar Roy Chowdhury**, Optics Letters, **45**, 3386 (2020)
56. *Multifold coupling Enabled High Quality factor Toroidal resonances in Terahertz Metasurfaces*, Deepak Kumar, Surya Pranav Ambatipudi, Sabyasachi Banerjee, Ranjan Kumar, and **Dibakar Roy Chowdhury**, Journal of Applied Physics, **127**, 193103 (2020)
57. *Strong terahertz Matter Interaction induced Ultrasensitive Sensing in Fano Cavity based Metamaterials*, Subhajit Karmakar, Deepak Kumar, Ravendra Varshney, and **Dibakar Roy Chowdhury**, Journal of Physics D: Applied Physics, **53**, 415101 (2020); doi.org/10.1088/1361-6463/ab94e3
58. *Bandwidth Enhancement of Planar Terahertz Metasurfaces via overlapping of dipole modes*, Deepak Kumar, Sabyasachi Banerjee, Ranjan Kumar, and **Dibakar Roy Chowdhury**, Plasmonics **15**, 1925 (2020) DOI: 10.1007/s11468-020-01222-7
59. *Polarization Independent double band electromagnetically Induced Transparency effect in terahertz metamaterials*, Rakesh Sarkar, K Monica Devi, Dipa Ghindani, Shriganesh Prabhu, **Dibakar Roy Chowdhury**, and Gagan Kumar, Journal of Optics, **22**, 035105 (2020)
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25. *Characterization of TbAs nanoparticles embedded in GaAs using pump probe measurements of carrier relaxation dynamics*, Laura Vanderhauf, Abul Azad, **Dibakar Roy Chowdhury**, Corey Bomberger, Joshua Zide, and Matthew Doty, APS March Meeting, Baltimore, Maryland, USA (2013)
26. *Metamaterials for broadband linear polarization conversion and near-perfect anomalous reflection and transmission*, Nathaniel Grady, Jane Heyes, **Dibakar Roy Chowdhury**, Yong Zeng, Antoinette Taylor, Diego Dalvit, Hou-Tong Chen, 4th International Conference on Metamaterials, Photonic Crystals and Plasmonics (META 2013), Sharjah, UAE (2013)

27. *Dark Plasmonic Modes Excitation through Symmetry Breaking*, Xiaofang Su, **Dibakar Roy Chowdhury**, Yong Zeng, Xiaoshuang Chen, and Abul Azad, META'14, Singapore (2014)
28. *Resonance Modulations in Metamaterials*, **Dibakar Roy Chowdhury**, Indian Innovations in Materials Research, Calcutta, India (2015)
29. *Terahertz guided mode propagation in a planar plasmonic waveguide and slow light properties*, Maidul Islam, Dibakar Roy Chowdhury, and Gagan Kumar, International Conference on Fibre Optics and Photonics, IIT Kanpur (2016)
30. Kumar, D., Rao, S. J. M., Kumar, G., & **Roy Chowdhury, D.** (2016, December). *Modulating the Fundamental Resonances in Near Field Coupled Planar Terahertz Metamaterials*. In International Conference on Fibre Optics and Photonics (pp. W4A-1). Optical Society of America; <https://doi.org/10.1364/PHOTONICS.2016.W4A.1>
31. *Near Field Inductive Coupling in Terahertz Metamaterials*, S Jagan Mohan Rao, Gagan Kumar, and **Dibakar Roy Chowdhury**, JSAP-OSA Joint symposia, Tokyo, Japan (2017)
32. C. S. Amith, S. Banerjee, E. A. Reddy, M. Latnekar and **D. Roy Chowdhury**, "Dark Mode Based Thin Film Sensing in Terahertz Metamaterials," *IEEE Workshop on Recent Advances in Photonics (WRAP)*, Hyderabad, 2017, pp. 1-3, doi.10.1109/WRAP.2017.8468546
33. *Electromagnetic response of dielectric nanostructures*, S Amarganti, **D Roy Chowdhury**, M Ratvik, and J Dontabhaktuni, Emerging Liquid Crystal Technologies, SPIE Photonics West, San Francisco, USA (2018)
34. *Tuning near field capacitive coupling in planar terahertz metamaterials*, S Jagan Mohan, Maidul Islam, Gagan Kumar and **Dibakar Roy Chowdhury**, SPIE Photonics West, San Francisco, USA (2018)
35. *Controlling extra ordinary transmission through hole arrays using sub wavelength periodic structures*, N Lok Abhishikht, S Banerjee, S Rangu, S Kamireddy, A K Azad, and **Dibakar Roy Chowdhury**, OSA Photonics Congress, Zurich, Switzerland (2018);
36. *Ultrafast Switching in Single Resonator based Metamaterials*, S Jagan Mohan Rao, Gagan Kumar, and **Dibakar Roy Chowdhury**, OSI International Symposium in Optics, IIT Kanpur, Kanpur (2018)
37. *Electrically controlled Terahertz Modulator based on Fano Metamaterials*, S. Karmakar, R. K. Varshney, **D. Roy Chowdhury**, *International Conference in Optics Photonics, SP137*, ISBN: 978-93-88653-41-1, IIT Delhi, Delhi, India (2018)
38. *Localized Magnetic Field Enhancement in Fano Resonance based Active Metamaterials*, Subhajit Karmakar, Ravi Varshney, and **Dibakar Roy Chowdhury**, pp. 306 – 307, OSI International Symposium in Optics, IIT Kanpur, Kanpur (2018)
39. *Broadband Terahertz Planar Metamaterials*. Kumar, D., Kumar, R., & **Roy Chowdhury, D.** The International Conference on Fiber Optics and Photonics (PHOTONICS 2018), Indian Institute of Technology Delhi, New Delhi, India TP 137, ISBN: 978-93-88653-41-1
40. *Thin Film Sensing with Asymmetric Terahertz Metasurfaces*, S. Banerjee, D. Kumar, S. Goel, B. P. Pal and **D. Roy Chowdhury**, *2019 Workshop on Recent Advances in Photonics (WRAP)*, Guwahati, India, 2019, pp. 1-3

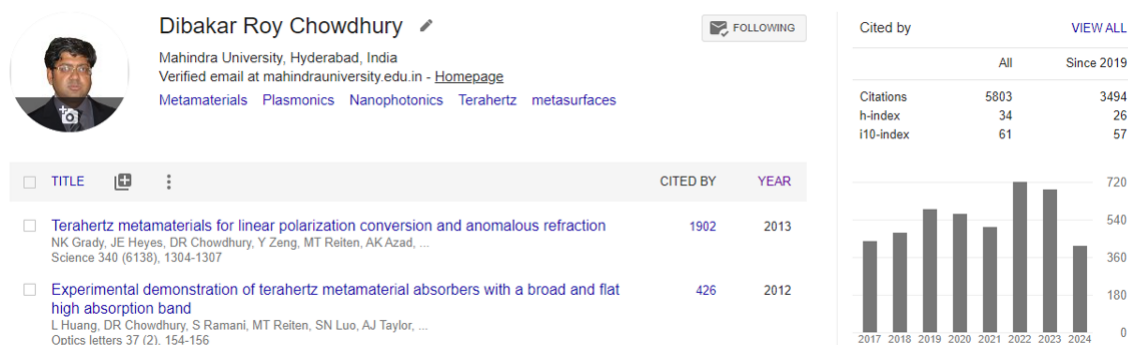
41. *Displacement sensing using terahertz metasurfaces*, S. Karmakar, R. K. Varshney, and **D. Roy Chowdhury**, presented in *Workshop on Recent Advances in Photonics (WRAP)*, paper id: **SP-48**, held at IIT Guwahati, Dec.13-14, 2019. DOI: [10.1109/WRAP47485.2019.9013671](https://doi.org/10.1109/WRAP47485.2019.9013671)
42. *Polarization sensitive Terahertz modulator based on electrically controlled Fano metasurfaces*, S. Karmakar, R. K. Varshney, and **D. Roy Chowdhury**, *International Conference on Optics & Electro-optics (ICOL-2019)*, paper id: **TMM-PP-03**, Instruments R&D Establishment (IRDE), Dehradun, India, Oct.19-22, 2019. (Awarded as 'SPIE Best paper award')
43. *Controlling Resonance nature and Spectral Line shapes in Terahertz Metamaterials*, S. Karmakar, R. K. Varshney, and **D. Roy Chowdhury**, *OSA Frontiers in Optics/ Laser Science APS/DLS (FIO/LS 2019)*, **JW3A.74**, Washington DC, USA, 2019. DOI: <https://doi.org/10.1364/FIO.2019.JW3A.74>.
44. *Lattice induced strong coupling in capacitance loaded Terahertz Metasurfaces*, S. Karmakar, R. K. Varshney, and **D. Roy Chowdhury**, *Optics and Photonics: Theory & Computational Technique (OPTCT 2019)*, IIT Delhi, Delhi, India, 2019.
45. *Magnetic Near-field enhancement in THz Multilayer Fano Metamaterial*, S. Karmakar, R. K. Varshney, and **D. Roy Chowdhury**, *Conference on Lasers and Electro-Optics Europe and European Quantum Electronics Conference (CLEO/ EUROPE- EQEC)*, **eh_p_14**, Munich, Germany, 2019. URL: https://www.osapublishing.org/abstract.cfm?uri=EQEC-2019-eh_p_14

CITATIONS

Number of citations: 5803

h-index: 34

i10-index: 61



External Funds Successful:

Project Title: “Exploring exceptional points in terahertz metasurfaces and subsequent development of a quantum material sensor”

Core Research Grant (CRG) (2024)

Funding agency: SERB, Department of Science and Technology (DST), India

Amount Funded: 49 L (as PI)

Project Title: “Metamaterials Inspired Slow-light photonics for information and sensing related applications”

SERB-Science and Technology Award for Research (STAR) (2023)

Funding agency: Science and Engineering Research Board, India

Amount Funded: 38.80 L (**as PI**)

Project Title: “Harnessing non-linearity in metamaterials for realizing a high-power terahertz source”

Core Research Grant (CRG) (2020)

Funding agency: Department of Science and Technology (DST), India

Amount Funded: 25.01 L (**as PI**)

Project Title: “Metamaterials based compact broadband terahertz modulator for Terahertz photonics” (2021)

Funding agency: Ministry of Electronics and Information Technology (MeitY), India

Amount Funded: 74 L (approx) (with IIT Guwahati)

Project Title: “Development of Terahertz Metagratings for detection of ultra-small deep subwavelength particles” (2019)

Funding Agency: Board of Research in Nuclear Sciences (BRNS), Department of Atomic Energy, India

Amount Funded: 30 L (**as PI**)

Project Title: “Metamaterials for phase control of Resonant Tunneling Diode based oscillators”

Indo Austria International Research Funding Scheme (2018)

Funding agency: Department of Science and Technology (DST), India

Amount Funded: 16.4 L (**as PI**)

Project Title: “Electromagnetic Metamaterials based ultra thin film sensors”

Extra Mural Research (EMR) (2016)

Funding agency: Department of Science and Technology (DST), India

Amount Funded: 37.86 L (**as PI**)

Project Title: “Metamaterials induced perfect absorption in thin film absorber”

Early Career Research Award (2016)

Funding agency: Department of Science and Technology (DST), India

Amount Funded: 29.68 L (**as PI**)

Project Title: “Complex Dielectric Structures as Tunable Metamaterials”

Extra Mural Research (EMR) (2017)

Funding agency: Department of Science and Technology (DST), India

Amount Funded: 24.45 L (**as Co-PI**)

Project Title: “Developing perovskite solar cell using multiferroic $\text{Bi}_{1-x}\text{M}_x\text{FeO}_3$ (M=Pb, Ca, La) thin films”

Indo-Taiwan S&T Cooperation programme (2018)

Funding Agency: DST (India) & MOST (Taiwan)

Amount Funded: 25.0 L (approx) (**as Co-PI**)

Funds to conduct conference:

Project: IEEE WRAP 2017 (Conference in Photonics, 17 – 19 December, 2017)

Amount Funded: 8000.00 USD (5.2 L INR)

Funding Agency: IEEE Photonics Society

Internal Funds Received:

Project: Development of Photo lithography facility

Funding agency: Mahindra Ecole Centrale

Amount Funded: 130 L

Project: Development of High Frequency Laboratory

Funding agency: Mahindra Ecole Centrale

Amount Funded: 250 L

Project: Development of Metamaterials based devices

Funding agency: Mahindra Ecole Centrale

Amount Funded: 5 L

Merchandising:

Received order for 6 L INR to supply terahertz filters and polarizers to ACRHEM, University of Hyderabad, Hyderabad, India through a competitive purchase procedure

Invited Talks

1. *Terahertz spectroscopy and its applications*, **Dibakar Roy Chowdhury**, Europhotonics Consortium, Tampere University, Helsinki, Finland (2024)
2. *Transverse magnetism in 1-dimensional plasmonic system*, **Dibakar Roy Chowdhury**, iSNOE2, Shiv Nadar University, India (2024)
3. *Terahertz metamaterials and metasurfaces*, **Dibakar Roy Chowdhury**, SCAN lecture series, University of Hyderabad, India (2023)
4. *Magnetically tunable Terahertz metamaterials*, **Dibakar Roy Chowdhury**, OSA Photonics, IISc Bangalore, India (2023)
5. *Multi-layer Al/Ni/Al based plasmonic structures for Terahertz devices*, **Dibakar Roy Chowdhury**, ICMAT Singapore, (2023)
6. *Terahertz magnetism in low dimension plasmonic metamaterials*, **Dibakar Roy Chowdhury**, CEFIPRA workshop, CSIR-CSIO, Chandigarh, India (2023)

7. *Terahertz Sensing and Spectroscopy*, **Dibakar Roy Chowdhury**, Tutorial in IEEE ICEE, IISc Bangalore, India (2022)
8. *Terahertz Metamaterials I and II*, **Dibakar Roy Chowdhury**, Resource talk, HRDC, University of Hyderabad, India (2022)
9. *Transverse Terahertz Magnetism in 1-dimensinal Plasmonic system*, **Dibakar Roy Chowdhury**, URSI-RCRS 2022, IIT Indore, India (2022)
10. *Terahertz magnetism in low dimensional plasmonic systems*, **Dibakar Roy Chowdhury**, Ultrafast Sciences 2022, IISER Thiruvananthapuram, India (2022)
11. *Magnetism in 1-dimensinal Plasmonic Terahertz system*, **Dibakar Roy Chowdhury**, IEEE photonics society, International Day of Light (IDL), IIT Bombay, India (2022)
12. *Terahertz metamaterials*, **Dibakar Roy Chowdhury**, Frontiers in Terahertz Technologies and Applications (FTTA-2021), National Physical Laboratory (NPL), New Delhi, India (2021)
13. *Terahertz metamaterials and metasurfaces*, **Dibakar Roy Chowdhury**, Raman Optronics webinar series (ROWS-2021), University of Kerala, Trivandrum, India (2021)
14. *Robust THz Transmissions in Photonic Topological Insulators*, **Dibakar Roy Chowdhury**, Frontiers on Optics, Optical Society of India (OSI), IIT Delhi, Delhi, India (2021)
15. *Ultrafast Lasers: Need and potential applications*, **Dibakar Roy Chowdhury**, NTTR Chandigarh, faculty development program (FDP) (2021)
16. *Terahertz magnetism in 1-dimensinal wire*, **Dibakar Roy Chowdhury**, DLS workshop, IUCA, Delhi, India (2021)
17. *Terahertz Metamaterials*, **Dibakar Roy Chowdhury**, MLR Institute of Technology, Hyderabad (2020)
18. *Ultrafast Lasers: Need and potential applications*, **Dibakar Roy Chowdhury**, NTTR Chandigarh, faculty development program (2020)
19. *Excitation of Fano resonances in symmetric terahertz metasurfaces*, **Dibakar Roy Chowdhury**, SPIE Defence and Commercial Sensing, Annheim, USA (2020)
20. *Fano Resonances in Terahertz Metamaterials*, **Dibakar Roy Chowdhury**, Advanced Photonics Technology organized by CSIO, Chandigarh, India (2019)
21. *Near Field Coupling in Asymmetric Split Ring Resonators based metasurfaces*, **Dibakar Roy Chowdhury**, SPIE Defence and Commercial Sensing, Baltimore USA (2019)
22. *Passive and active control of resonant modes in Terahertz Metamaterials*, **Dibakar Roy Chowdhury**, Advances in Optics and Photonics, University of Hyderabad, Hyderabad, India (2019)
23. *Concentric Resonators based Broadband Terahertz Metasurfaces*, **Dibakar Roy Chowdhury**, Asia Pacific Radio Science Conference, New Delhi, India (2019)
24. *Resonance modulations in Terahertz Metamaterials*, **Dibakar Roy Chowdhury**, Photonics, Metamaterials and Plasmonics 2019, Jaypee University, Noida, India (2019)
25. *Active Terahertz Metamaterials*, **Dibakar Roy Chowdhury**, OSA PHOTONICS 2018, IIT Delhi, India (2018)
26. *Near Field Electromagnetic Coupling in Terahertz Metamaterials*, **Dibakar Roy Chowdhury**, Grand NanoPhotonics (GNP) Conference, TIFR, Mumbai, India (2018)

27. *Exploring the Near Field Inductive and Capacitive Coupling in Terahertz Metamaterials*, **Dibakar Roy Chowdhury**, TEQIP-III, Jorhat Engineering College, Assam, India (2018)
28. *Metamaterials for Terahertz Applications*, **Dibakar Roy Chowdhury**, IEEE Experts Lecture, SAC, ISRO, Ahmedabad, India (2018)
29. *Thin Film Sensing with Terahertz Metamaterials*, **Dibakar Roy Chowdhury**, Optical Society of India (OSI) Conference, IIT Kanpur, Kanpur, India (2018)
30. *Active and Passive Tuning of Terahertz Metamaterials Responses*, **Dibakar Roy Chowdhury**, University Lecture, Technical University of Vienna, Vienna, Austria (2018)
31. *Strongly near field coupled electromagnetic metamaterials*, **Dibakar Roy Chowdhury**, Optical Society of India (OSI) Conference, Hissar, Haryana, India (2017)
32. *How to publish research papers in high impact journals*, **Dibakar Roy Chowdhury**, IEEE Young Professionals Meeting, IIIT Hyderabad, Hyderabad (2017)
33. *Interactions of Terahertz Radiations with Metamaterials*, **Dibakar Roy Chowdhury**, Terahertz Spectroscopy, TEQIP, IIT Kharagpur (2017)
34. *Effect of resonator displacements in inductively coupled Terahertz Metamaterials*, **Dibakar Roy Chowdhury**, URSI Regional Conference on Radio Science (URSI-RCRS), Tirupati, India (2017)
35. *Metamaterials based Photonic Structures for Terahertz wavelengths*, **Dibakar Roy Chowdhury**, TEQIP, Andhra University, Vizag (2017)
36. *Near field coupled Terahertz Metamaterials*, **Dibakar Roy Chowdhury**, PHOTONICS 2016, IIT Kanpur, Kanpur, India (2016)
37. *Metamaterials based Photonics Structures for Terahertz*, **Dibakar Roy Chowdhury**, Advancement and applications of ultrashort Laser pulses, TEQIP II, IIT Guwahati (2016)
38. *Resonance Modulations in Electromagnetically Coupled Terahertz Metamaterials*, **Dibakar Roy Chowdhury**, IEEE Optical MEMS and Nanophotonics (IEEE OMN), NUS, Singapore (2016)
39. *Near Field Coupled Terahertz Metamaterials*, **Dibakar Roy Chowdhury**, SPIE Terahertz Devices, Baltimore, USA (2016)
40. *Terahertz Metamaterials*, **Dibakar Roy Chowdhury**, Workshop on Terahertz Technology, IIT Delhi (2015)
41. *Near field coupled Metamaterials for Terahertz Wavelengths*, **Dibakar Roy Chowdhury**, Conference on Computational Physics (CCP), IIT Guwahati (2015)
42. *Resonance Modulations in Terahertz Metamaterials*, **Dibakar Roy Chowdhury**, Terahertz Technology Current Scenario and Future Trends, Thirubananthapuram, India (2015)
43. *Ultrafast Active Terahertz Metamaterials*, **Dibakar Roy Chowdhury**, Central Glass and Ceramic Research Institute (CGCRI), Calcutta, India (2015)
44. *Ultrafast Active Terahertz Metamaterials*, **Dibakar Roy Chowdhury**, Molecular Materials Meetings (M3), Singapore (2014)
45. *Ultrafast Active Terahertz Metamaterials*, **Dibakar Roy Chowdhury**, Focussed Discussion Meeting on Metamaterials and Nanostructures, Kanpur, India (2013)

46. *High frequency behavior of Resonant Tunneling diode and modulation of resonances in terahertz metamaterials*, **Dibakar Roy Chowdhury**, Ruhr University, Bochum, Germany (2012)
47. *Resonant Tunneling diode and resonance modulation in split ring resonators*, **Dibakar Roy Chowdhury**, Indian Institute of Science, Bangalore, India (2011)
48. *Negative Differential conductance of Resonant Tunneling Diode at high frequencies*, **Dibakar Roy Chowdhury**, Technical University of Eindhoven, Eindhoven, Netherlands, (2008)

Other Activities

- Convener, Indo-French Conference on Frontiers in Photonics and Metamaterials, Mahindra University, India (2023)
- Topical Editor, Optics Letters, Optica (erstwhile OSA) (2022 -)
- Associate Editor in “Frontiers in Physics” (2022 -)
- Technical Track Chair: Terahertz Photonics, IEEE WRAP 2022, IIT Bombay, India (2022)
- Co-Chair, International Conference on Technologies for Smart Green connected Society (ICTSGS, [ICTSGS](#)), strongly backed by United Nations, (2021)
- Review Editor in “Frontiers in Physics – Optics and Photonics” (2021 - 2022)
- Panelist in Photonics Vertical, Vaishwik Bharatiya Vaigyanik (VAIBHAV) Summit, Government of India, (2020)
- Leader, Nanotechnology program, Mahindra University, (2020 - ongoing)
- Head of Physics Department, Mahindra University, (2019 - 2024)
- Technical Program Committee Chairman, IEEE WRAP 2019, IIT Guwahati (2019)
- Session Chair, Terahertz Spectroscopy, SPIE Defense and Commercial Sensing 2019, Baltimore, USA (2019)
- Technical Program Committee member OSA PHOTONICS 2018, IIT Delhi, India (2018)
- Technical Program Committee (TPC) member of Optical Society of India (OSI) research symposium (2018)
- Convener, IEEE Workshop on Recent Advancements in Photonics (WRAP-2017)
- Faculty in charge, Sports, MEC (2017 - 2020)
- Coordinator of Physics, Mahindra Ecole Centrale (2017 - 2019)
- Visiting Professor, LPQM, Centrale Supélec Paris, France (2016)
- Advisory Committee Member OPTRONIX organized by UEM Calcutta (OPTRONIX-2016)
- Technical Program Committee (TPC) member of IEEE Workshop on Recent Advancements in Photonics (WRAP-2015)
- Proposal reviewer for Prime Minister Research Fellowship (PMRF), Government of India
- Proposal reviewer for DST, India, Nanyang Technological University, Singapore, Tier-I & Tier II proposals, National Science Center, Poland, Distinguished Proposal Review Committee, CINT, Sandia National Laboratory, USA

- PhD Thesis Reviewer, Nanyang Technological University, Singapore, University of Hyderabad (UoH), Hyderabad, RMIT Melbourne, IIST Trivandrum, IIT Guwahati, IIT Roorkee etc.

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